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```
> bootcamp.learn()
```

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JomCode Coding Bootcamp

CHAPTER 06

Loops and data processing

```
learn / type / run / explain
```

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Plain meaning and technical explanation

PLAIN MEANING

Repeat an action for each item instead of writing it repeatedly.

TECHNICAL EXPLANATION

for iterates over an iterable. while repeats while a condition is true. Comprehensions build collections from iteration.

REAL-WORLD USE

List posts, filter results and calculate totals.

QUICK TIP

Keep loop bodies small and make while loops progress toward stopping.

INSTALL / RUN

No packages. Windows: py workbook.py. macOS: python3 workbook.py.

Worked example 1: Number each post

workbook.py

```
titles = ["Hello", "Launch"]
for number, title in enumerate(titles, start=1):
    print(f"{number}. {title}")
```

PRO TIP Prefer iteration over indexing when you need the value.

Run and explain example 1

EXPECTED RESULT

1. Hello
2. Launch

WHY IT BEHAVES THIS WAY

enumerate() supplies a counter and the current value.

Worked example 2: Filter and transform

workbook.py

```
posts = [{"title": "Hello", "views": 2},
         {"title": "Launch", "views": 12}]
popular = [p["title"].upper() for p in posts
           if p["views"] >= 10]
print(popular)
```

PRO TIP Use an ordinary loop if a comprehension becomes hard to read.

Run and explain example 2

EXPECTED RESULT

['LAUNCH']

WHY IT BEHAVES THIS WAY

The condition selects matching records before the expression creates output values.

Worked example 3: A bounded while loop

workbook.py

```
01    attempt = 0
02    while attempt < 3:
03        attempt += 1
04        print(f"Attempt {attempt}")
```

PRO TIP Real API retries also need delay and retryable-error checks.

Run and explain example 3

EXPECTED RESULT

Attempt 1
Attempt 2
Attempt 3

WHY IT BEHAVES THIS WAY

The counter increases each time, eventually making the condition false.

Worked example 4: Totals and early exit

workbook.py

```
views = [3, 0, 7]
total = 0
for value in views:
    total += value
print(total)
for value in views:
    if value == 0:
        print("Unviewed post found")
        break
```

PRO TIP sum(views) is a concise alternative for a numeric total.

Run and explain example 4

EXPECTED RESULT

10
Unviewed post found

WHY IT BEHAVES THIS WAY

The accumulator keeps a running total. break exits the nearest loop.

Exercise 1: Search titles

YOUR TASK

Return titles containing the search text, ignoring case.

CHECK THESE CASES

"py" matches Python Tips. Empty search matches every title.

HINT AFTER A TRY

Normalize both strings with lower().

Exercise 2: Author counts

YOUR TASK

Count posts per author using a dictionary.

CHECK THESE CASES

Ada, Lin, Ada gives Ada=2 and Lin=1. Empty input produces {}.

HINT AFTER A TRY

Use counts.get(author, 0) + 1.

Checkpoint before moving on

01 Run and explain all four worked examples. Match their actual results to the expected behavior.

02 Attempt both exercises before reading the solutions. Verify every acceptance case.

03 Explain how this chapter supports Postlight: List posts, filter results and calculate totals.

Official reference